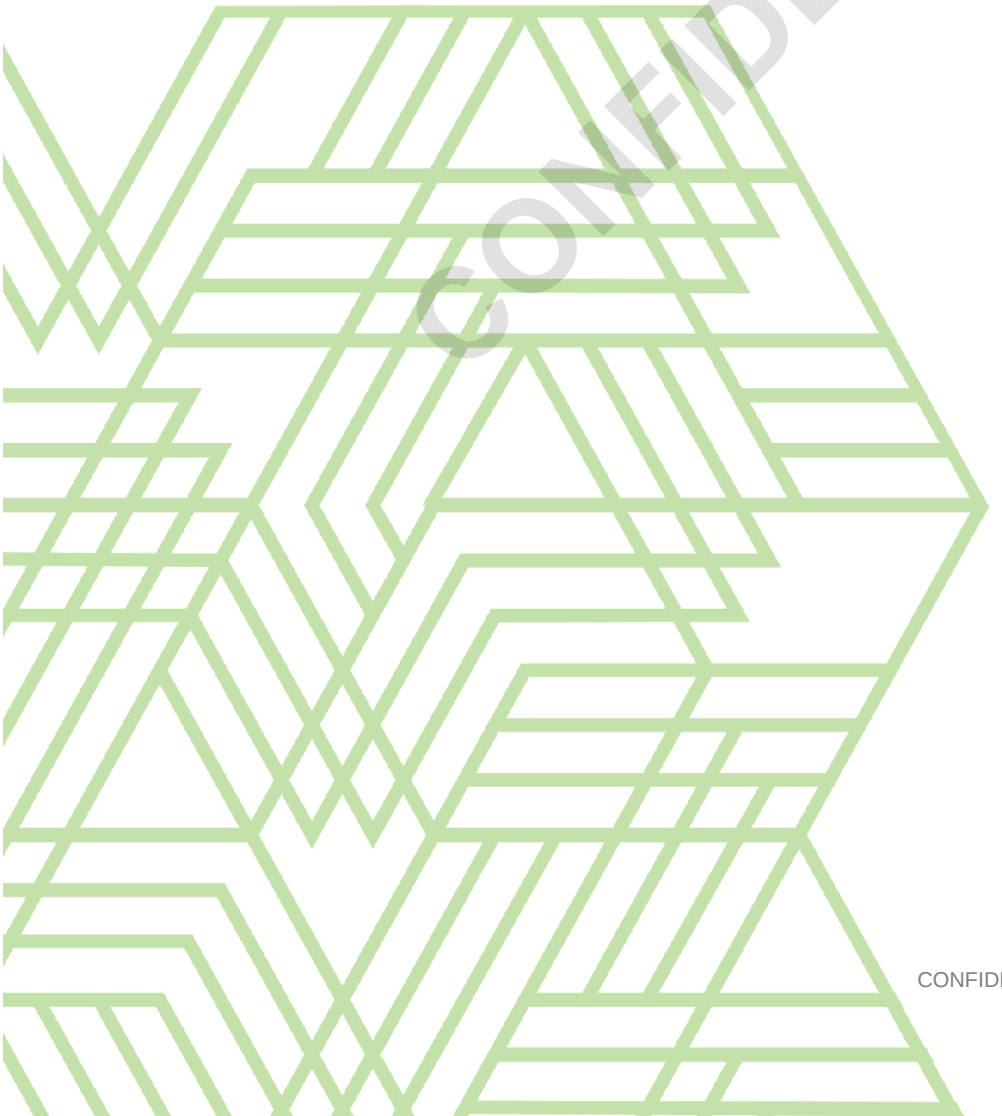
ATM33/e Support for External Energy Storage Devices

Application Note

SUMMARY: This application note describes the functionality of the ATM33/e Wireless SoC Series EVK operating from a high-voltage battery or storage device, and the ATM33e charging performance from PV or RF harvesting.

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Table of Contents

Acronyms and Abbreviations 3

1. Overview 4

2. Hardware and Software Requirements 4

2.1 EVKs for Energy Harvesting or Li-ion Battery 4

2.2 SoCs for Energy Harvesting or Li-ion Battery 5

2.3 Supported SDK 5

3. Power Management and Harvesting 5

3.1 VBATLI Supply Path Operation for ATM33 and ATM33e 7

3.2 VBATLI Charging Path Operation for ATM33e 10

3.3 VBATLI System Brownout for ATM33 and ATM33e 13

4. Battery Charging Performance 14

4.1 PV Harvesting 14

4.2 RF Harvesting 15

5. Software Settings 16

5.1 Enable VBATLI for ATM33 16

5.2 Enable VBATLI for ATM33e 17

5.3 Modify VBATLI Brownout Levels 20

5.4 Battery Monitoring for VBATLI 20

Reference Documents 21

Revision History 21

List of Figures

Figure 1 - Block Diagram for ATM33 Operating from VBATLI

Figure 2 - Block Diagram for ATM33e Operating from VBATLI

Figure 3 - Operation Diagram for VBATLI Supply Path

Figure 4 - VBATLI Bypass Behavior from High Voltage to Low

Figure 5 - VBATLI Bypass Behavior from Low Voltage to High

Figure 6 - VBATLI Charging Path 1: Charging Through An External Charge Pump for ATM33e

Figure 7 - VBATLI Charging Path 2: Charging Through On-chip Path for ATM33e

CONFIDENTIAL

Figure 8 - Battery Charging Operations for Different Battery Levels

Figure 9 - PV Charging Efficiency Through the External Charge Pump

Figure 10 - PV Charging Efficiency Through the On-chip Path

Figure 11 - RF Charging Efficiency

List of Tables

Table 1 - EVKs for Harvesting or Li-ion Battery

Table 2 - SoCs for Harvesting or Li-ion Battery

Table 3 - VBATLI LDO Related Thresholds Descriptions

Table 4 - VBATLI Charging Related Thresholds and Descriptions

Table 5 - VBATLI Brownout Related Thresholds and Descriptions

Acronyms and Abbreviations

Acronyms	Definition
ATM33	ATM3325 ATM3330
ATM33e	ATM3330e
ATM33/e	ATM3325 ATM3330 ATM3330e
EVK	Evaluation Kit
PV	Photovoltaic
RF	Radio Frequency
SDK	Software Development Kit

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1. Overview

This application note describes the functionality of the ATM33 and ATM33e SoC operating from a high-voltage battery or storage device (i.e. Li-ion battery, LiFePO4 battery or hybrid super capacitor). This document also describes ATM33e harvester and its charging performance from PV or RF harvesting methods.

For the harvesting performance with a low-voltage non-rechargeable battery at VBAT, please refer to the **ATM33e Series Energy Harvesting Application Note**.

This document also includes programming instructions to demonstrate how to enable various battery and harvesting settings using the Atmosic SDK.

2. Hardware and Software Requirements

2.1 EVKs for Energy Harvesting or Li-ion Battery

EVK	SoC Package	SoC Part Number	Kit Part Number	Harvesting Support	Li-ion Support
Evaluation Kit for ATM3325	40-pin 5x5 mm QFN	ATM3325-5DCAQK	ATMEVK-3325-QN	Not Supported	Not Supported
Evaluation Kit for ATM3330	56-pin 7x7 mm QFN	ATM3330-5DCAQN	ATMEVK-3330-QN	Not Supported	**Supported**
Evaluation Kit for ATM3330e	56-pin 7x7 mm QFN	ATM3330E-5DCAQN	ATMEVK-3330e-QN	**Supported**	**Supported**

Table 1 - EVKs for Harvesting or Li-ion Battery

***Note**: Energy harvesting feature is applicable to ATM33e Wireless SoC series only.*

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2.2 SoCs for Energy Harvesting or Li-ion Battery

SoC Part Number	SoC Package	Harvesting Support	Li-ion Support
ATM3330E-5DCAQK	40-pin 5x5 mm QFN	**Supported**	**Supported**
ATM3330E-5DCAQN	56-pin 7x7 mm QFN	**Supported**	**Supported**
ATM3330-5DCAQN	56-pin 7x7 mm QFN	Not Supported	**Supported**
ATM3325-5DCACM	49L WLCSP	Not Supported	**Supported**
ATM3325-5DCAQK	40-pin 5x5 mm QFN	Not Supported	Not Supported

Table 2 - SoCs for Harvesting or Li-ion Battery

2.3 Supported SDK

Please download the latest Atmosic SDK and release notes from Atmosic Support Portal. Refer to the SDK User Guide for more information.

For related documents, please refer to the Reference Documents section.

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3. Power Management and Harvesting

The VBATLI pin is used as the input for a high-voltage battery or storage device with input range up to 4.2 V. The block diagrams for VBATLI operations are shown in Figure 1 and Figure 2 for ATM33 (excludes ATM3325 5x5mm package) and ATM33e, respectively. Only the energy supply pins (VBATLI, VBAT, VHARV, VSTORE, HARV_INP, HARV_INN) are shown in these figures. For the rest of the PMU pin connections and the external board components, please refer to the **ATM33/e Series Reference Manual** for the block diagrams with VBATLI (see All Supported PMU Configurations). Note when VBATLI is used, the VBAT pin should only have an external capacitor and should not be driven externally.

There is an LDO that regulates the VBATLI voltage down to VBAT and then supplies to the rest of the system through the switching regulator. When the VBATLI voltage drops, this LDO can be bypassed through the switch S1. This operation will be discussed in section VBATLI Supply Path Operation for ATM33 and ATM33e.

For ATM33e that supports charging with harvesting power (see Figure 2), the input can either be PV harvesting from a PV cell connected to VHARV or RF harvesting collected

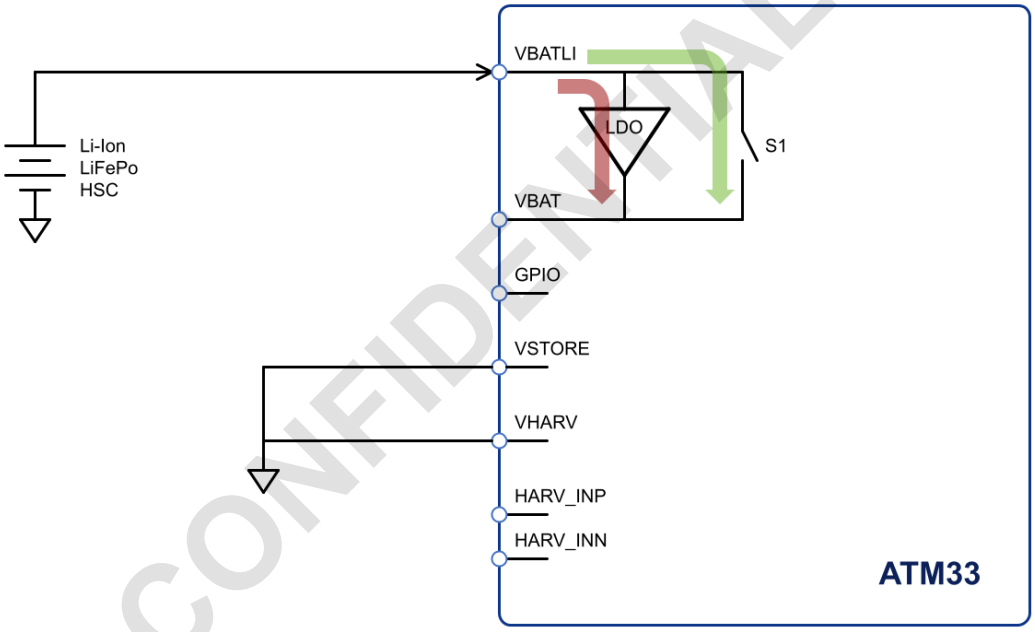
by the antenna connected at the HARV port of the board, then through the on-board matching network to HARV_INP and HARV_INN (connections not shown in Figure 2).

There are also two charging path options from the harvested power at VSTORE to the battery or storage device at VBATLI:

1) through the external charge pump boost circuit with the charge pump clock enabled through a GPIO

2) through the on-chip path with switcher S2 connected

This operation will be discussed in section VBATLI Charging Path Operation for ATM33e.



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Figure 1 - Block Diagram for ATM33 Operating from VBATLI

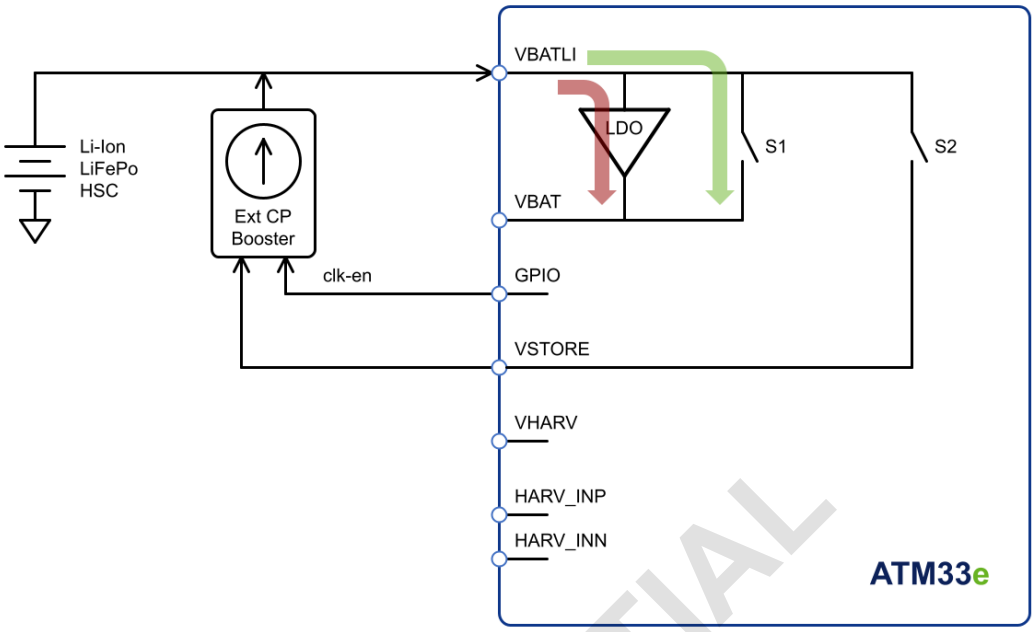


Figure 2 - Block Diagram for ATM33e Operating from VBATLI

When the battery or storage device level is too low at VBATLI, the system should be programmed to enter brownout. This feature is discussed in section VBATLI System Brownout for ATM33 and ATM33e.

3.1 VBATLI Supply Path Operation for ATM33 and ATM33e

The supply path from VBATLI can either go through an LDO to VBAT (see red arrow) or through an internal switch S1 to VBAT (see green arrow), as shown in Figure 1 and Figure 2 for ATM33 and ATM33e, respectively.

Figure 3 demonstrates the battery supply operations for different battery levels. There is an internal ADC (denoted as PMU ADC in Figure 3) that monitors VBATLI every 16ms during all supported power states. Depending on the voltage levels, the LDO and S1 path are controlled automatically by the hardware.

***Warning**: VBATLI has a maximum ramp rate as 15 V/s. Exceeding the maximum rate can potentially degrade the performance or permanently damage the device.*

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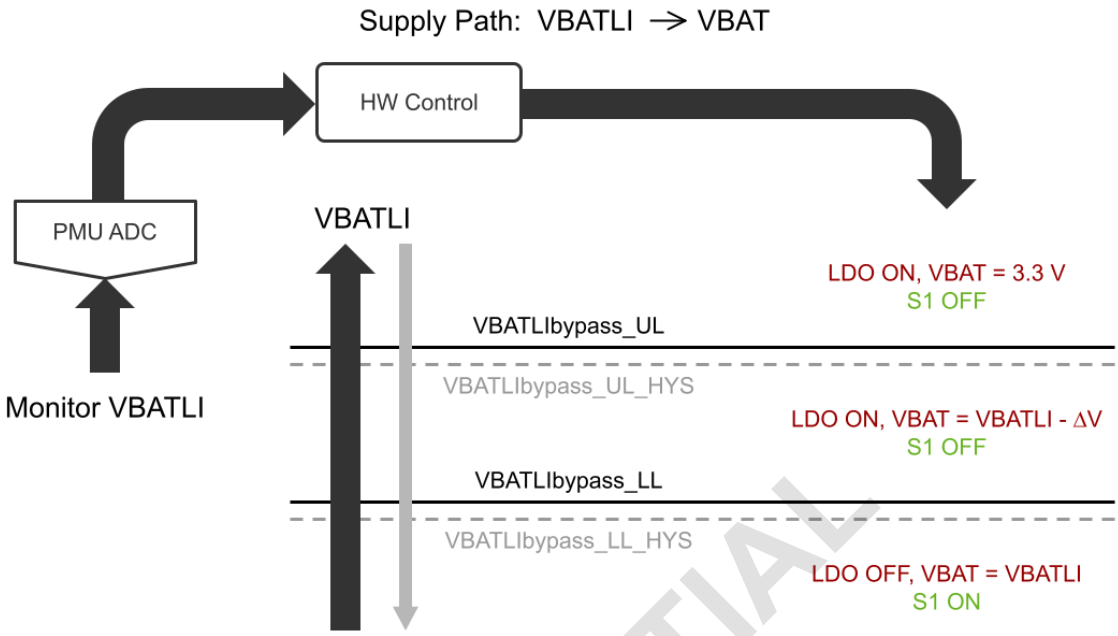


Figure 3 - Operation Diagram for VBATLI Supply Path

When VBATLI is high, above **VBATLIbypass_UL**, VBAT is regulated at **VBATldo_reg** by the LDO. When VBATLI is between **VBATLIbypass_UL** and **VBATLIbypass_LL**, VBAT moves along with VBATLI to optimize for current consumption. When VBATLI is below **VBATLIbypass_LL**, the LDO is turned off and the bypass switch S1 is ON. VBAT is at the same voltage as VBATLI. This operation is demonstrated in Figure 4 and Figure 5. The voltage thresholds are described in Table 3.

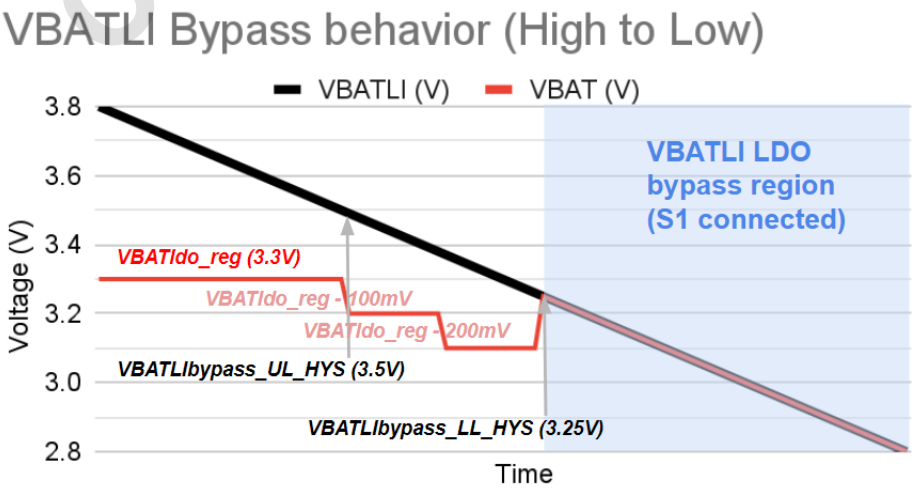


Figure 4 - VBATLI Bypass Behavior from High Voltage to Low

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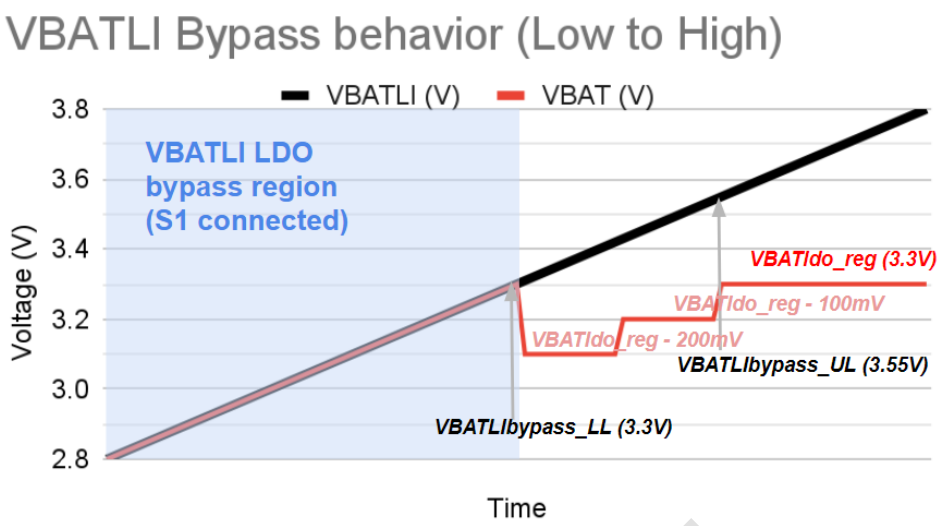


Figure 5 - VBATLI Bypass Behavior from Low Voltage to High

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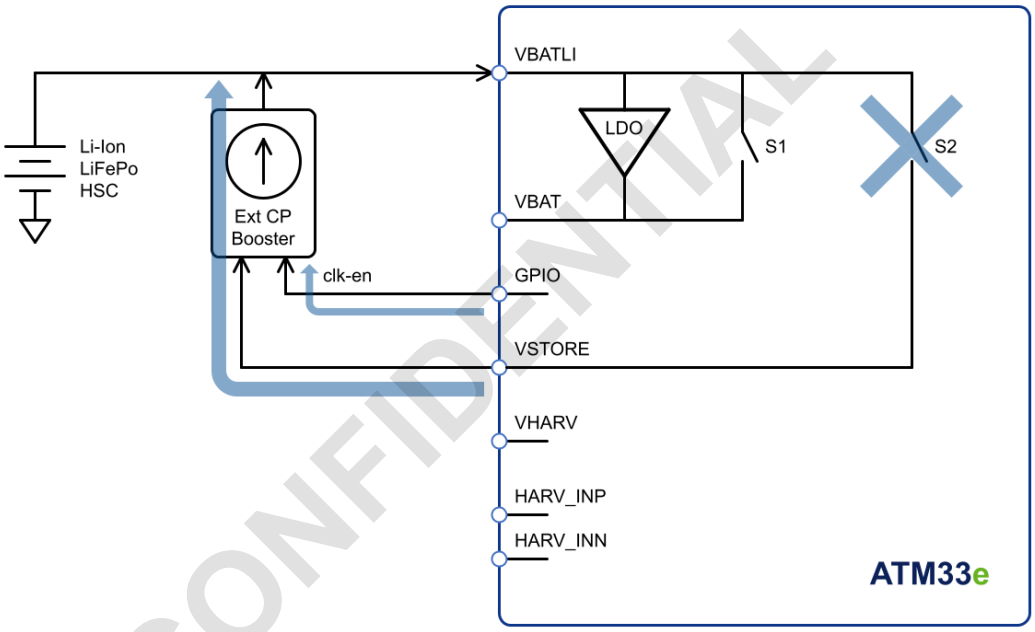
VBATLI LDO Related Thresholds	Descriptions	SW App Default Value	Typical Value
VBATLIbypass_UL	VBATLI upper level above which the VBAT is regulated at a constant voltage level (VBATldo_reg). This threshold is typically 250 mV above VBATLIbypass_LL.		3.55 V
VBATLIbypass_UL_HYS	Hysteresis level applies for the falling transition, typically 50 mV below VBATLIbypass_UL.		3.5 V
VBATLIbypass_LL	VBATLI lower level above which the VBAT is regulated by the LDO. (programmable)	3.3 V	3.3 V
VBATLIbypass_LL_HYS	Hysteresis level applies for the falling transition, typically 50mV below VBATLIbypass_LL.		3.25 V
VBATldo_reg	VBAT regulated voltage when VBATLI LDO is ON and above VBATLIbypass_UL. (programmable)	3.3 V	3.3 V

Table 3 - VBATLI LDO Related Thresholds and Descriptions

3.2 VBATLI Charging Path Operation for ATM33e

For ATM33e, there are two charging paths from the harvested power at VSTORE to the battery or storage device at VBATLI. One of the two paths is selected for charging efficiency optimization at different battery levels. They are:

1) charging through the external charge pump boost circuit with the charge pump clock enabled through a GPIO, shown in Figure 6
2) charging through the on-chip path with switcher S2 connected, shown in Figure 7



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Figure 6 - VBATLI Charging Path 1: Charging Through An External Charge Pump for ATM33e

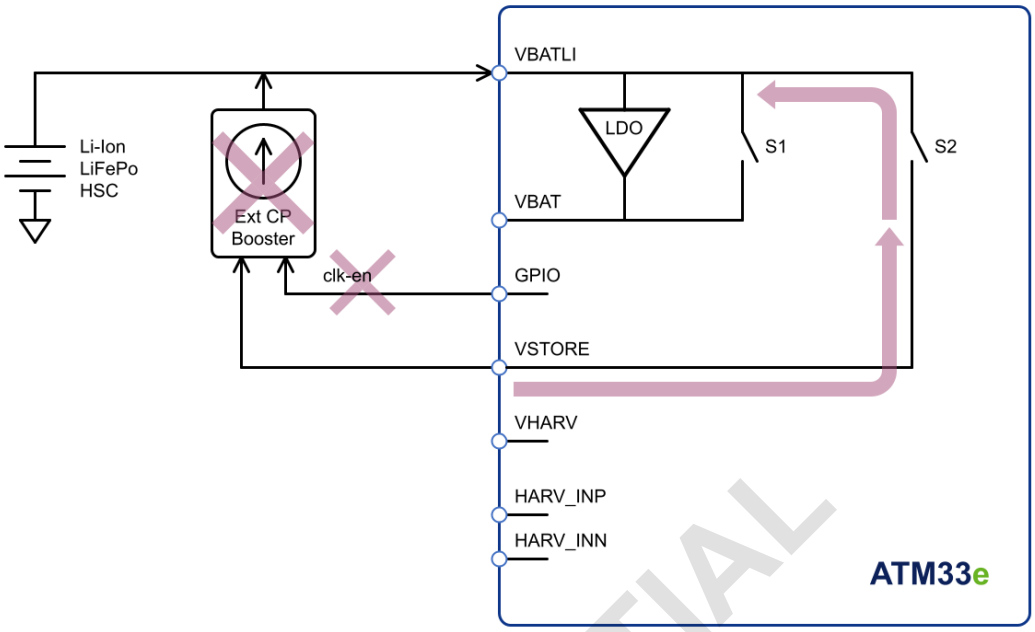


Figure 7 - VBATLI Charging Path 2: Charging Through On-chip Path for ATM33e

Figure 8 demonstrates the battery charging operations for different battery levels.

When the battery monitor feature is enabled in the SDK (refer to section Enable VBATLI for ATM33e), the software measures VBATLI periodically through GADC. Based on that voltage compared to the pre-defined programmable threshold levels, the software decides to enable or disable the two charging paths by controlling the charge pump clock (clk_en) and the internal switch S2.

Note: *The harvesting power is utilized for the load before it is used for battery charging at all times.*

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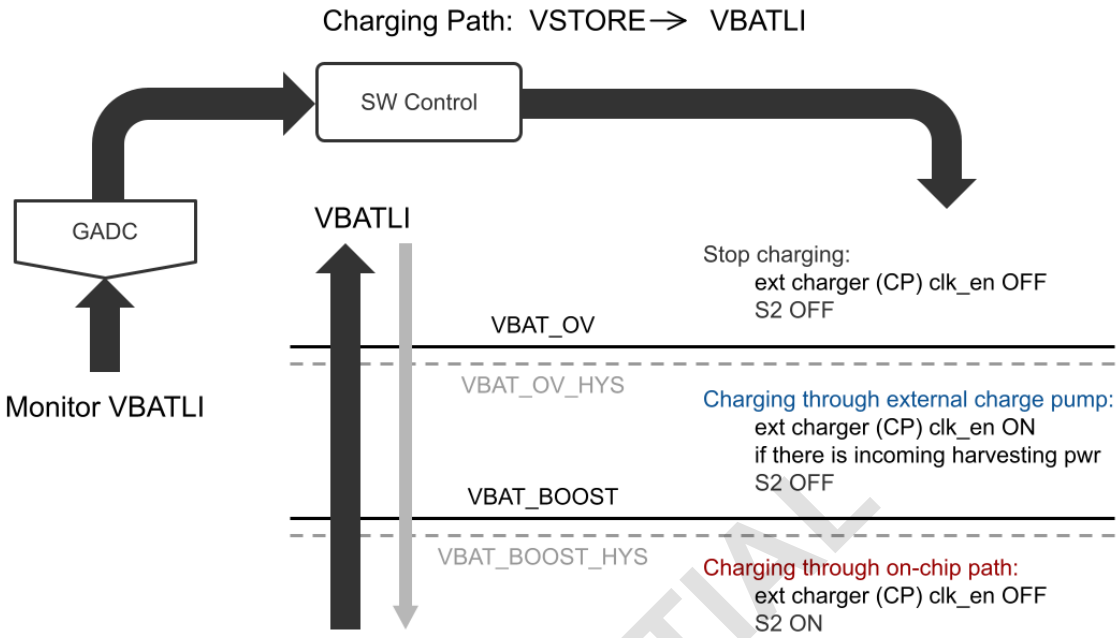


Figure 8 - Battery Charging Operations for Different Battery Levels

When VBATLI is high, above **VBAT_OV**, both VBATLI charging paths are disabled so there is no charging to VBATLI.

When VBATLI is between **VBAT_OV** and **VBAT_BOOST**, the charging path 1 is enabled by enabling the charge pump clock when there is enough harvested power on VSTORE.

When VBATLI is below **VBAT_BOOST**, the charging path 2 is enabled by connecting switcher S2 for on-chip charging and disabling the charge pump clock.

The threshold **VBAT_OV** is chosen based on the maximum spec of the battery or storage device being used. The programmable voltage thresholds are described in Table 4.

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VBATLI Charging Related Thresholds	Descriptions	SW Default Setting
VBAT_OV	Battery overvoltage level above which all battery charging should stop. This should be user-defined for different battery types.	3.8 V
VBAT_OV_HYS	Hysteresis applies for the falling transition. Recommend to set at 50 mV below VBAT_OV	3.75 V
VBAT_BOOST	Battery level above which the external boost charger operation is enabled when there is incoming harvesting power. This should be set lower than the max level VSTORE can charge up to (VSTORE_MAX). Recommend to set at 100 mV below the VSTORE_MAX.	3.2 V
VBAT_BOOST_HYS	Hysteresis applied for the falling transition. Recommend to set at 50 mV below VBAT_BOOST	3.15 V

Table 4 - VBATLI Charging Related Thresholds and Descriptions

***Warning**: the charging path 1 through the external charge pump boost circuit should not be turned on without an external storage device present at VBATLI pin.*

When the charge pump boost circuit is turned on and there is a storage device connected to the output of the booster (VBATLI pin), the storage device presents a low-impedance to the boost circuit, and thus the VBATLI voltage will not exceed the voltage of the storage device. When there is no storage device at VBATLI, the load impedance of the boost circuit is high and the resulting voltage on VBATLI pin can go up to V(VDDIO) + V(VSTORE_MAX) which is up to V(VDDIO) + 3.3V. The VBATLI pin is not tolerant to this level, and can result in permanent pin damage. Please refer to the **ATM33e Series Datasheet** for VBATLI pin rating.

Customers should take special care of this situation during evaluation or testing in manufacturing environments when storage devices may or may not be present.

3.3 VBATLI System Brownout for ATM33 and ATM33e

The brownout driver is enabled by default in the SDK. The software measures VBATLI periodically through GADC. When the battery or storage device voltage is too low at VBATLI, below **BROWNOUT_THR_VBATLI** level, the brownout signal is transitioned from low to high. If the chip is in active state when this situation happens, an interrupt is detected and the chip will be programmed to enter the lowest power state. If the chip is in sleep state when this situation happens, the chip will wake up first and then be

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programmed to enter the lowest power state. This is applicable for both ATM33 and ATM33e.

For ATM33e that allows battery charging, the internal PMU ADC monitors VBATLI voltage when the system is in brownout state. When the battery is charged above the exit level programmable by **WAKEUP_THR_VBATLI**, the chip exits brownout and reboot the application.

The voltage thresholds are described in Table 5. They are chosen based on the minimum spec of the battery or storage device being used at VBATLI.

VBATLI Brownout Related Thresholds	Descriptions	SW Default Setting
BROWNOUT_THR_VBATLI	Battery level to enter system brownout (VBATLI falling transition).	2.8 V
WAKEUP_THR_VBATLI	Threshold to set the battery level to exit system brownout (VBATLI rising transition). VBATLI = 1.8 + 0.05*X where 14 <= X <= 31	26 (corresponds to 3.1 V)

Table 5 - VBATLI Brownout Related Thresholds and Descriptions

Note 1: *The supported wakeup threshold is up to 3.35 V. Please contact Atmosic if your battery has a min spec higher than this level.*

Note 2: *For ATM33e, please make sure the wakeup threshold is set above the brownout threshold for at least 100 mV.*

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4. Battery Charging Performance

This section details the performance of the ATM3330e battery charging with PV and RF harvesting.

4.1 PV Harvesting

For PV harvesting, the PV cell is attached to the VHARV pin. The charging efficiency is measured from VHARV to the battery or storage device at VBATLI, which includes the VHARV to VSTORE efficiency and the VSTORE to VBATLI efficiency, either through the external charge pump circuit or directly through the connected switcher S2.

To evaluate charging efficiencies, hibernation is used as the chip load for demonstration. Note that the harvesting power is utilized first for the load before charging the VBATLI.

Figure 9 shows the PV charging efficiency through the external charge pump circuit. Figure 10 shows the PV charging efficiency through the on-chip path.

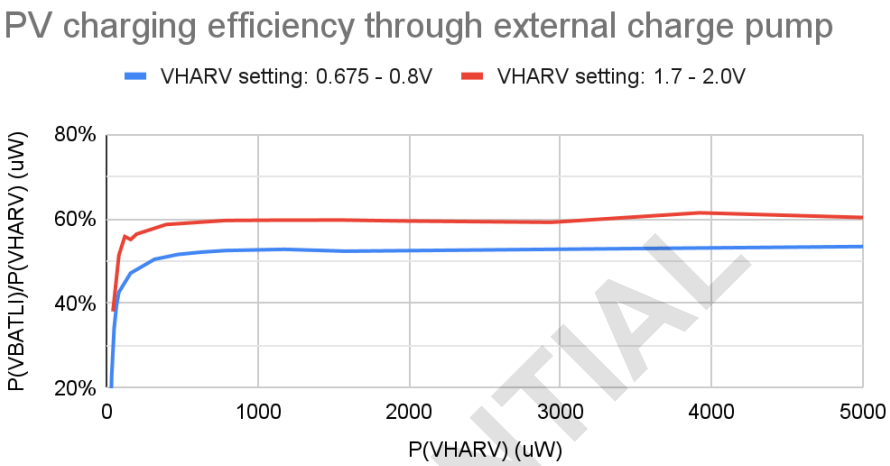


Figure 9 - PV Charging Efficiency Through the External Charge Pump

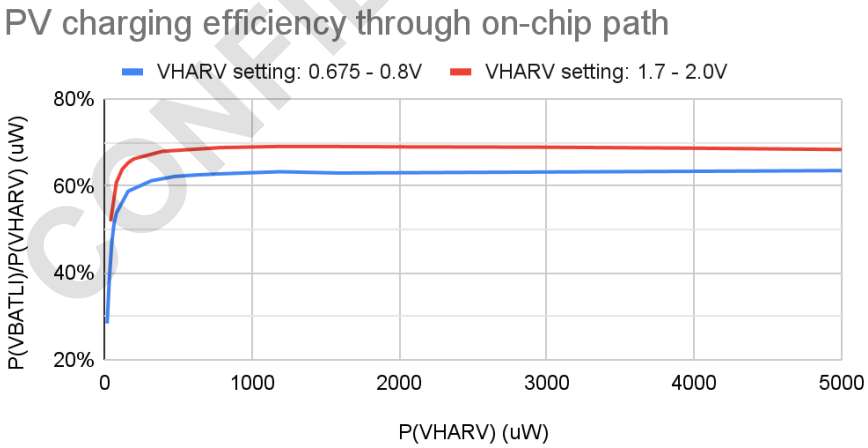


Figure 10 - PV Charging Efficiency Through the On-chip Path

4.2 RF Harvesting

For RF harvesting, the charging efficiency is measured from the received port of the board (HARV port) to the battery or storage device at VBATLI, which includes the on-board matching network efficiency, VHARV to VSTORE efficiency, and the VSTORE to

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VBATLI efficiency, either through the charge pump circuit or directly through the connected switcher S2.

To measure for charging efficiencies, hibernation is used as the chip load for demonstration. Note that the harvesting power is utilized for the load first before being used to charge VBATLI.

Figure 11 shows the RF charging efficiency either through the external charge pump circuit or through the on-chip path.

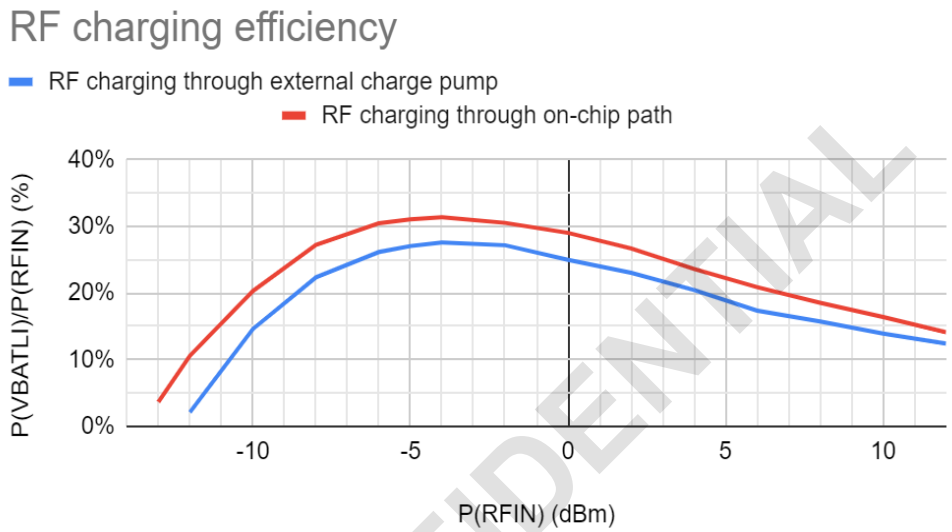


Figure 11 - RF Charging Efficiency

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5. Software Settings

The BLE_harv_adv application is pre-loaded on the ATM33e Evaluation Board. With the corresponding DIP switch settings on board, the user can evaluate beacon applications operated from a hybrid supercapacitor (max at 3.8 V) and its charging with RF or PV harvesting energy. Please refer to the **ATM33e Series Energy Harvesting EVK Quick Start Guide** for the corresponding DIP switch settings.

The following sections describe the software setting details for the relevant options.

5.1 Enable VBATLI for ATM33

To enable high-voltage battery or storage device on VBATLI in the SDK examples, the user can insert the following three lines into the CFLAGS in the corresponding makefile (ie. platform/atm33/ATM33xx-5/examples/BLE_adv/makefile), build the application and

load it to the target device.

```
CFLAGS += \
    -DBATT_TYPE=BATT_TYPE_LI_ION \
    -DBATT_LEVEL=BATT_LEVEL_GT_1P8V \
    -DVDDIO_TYPE=VDDIO_TYPE_INTERNAL \
```

Please note if the application is loaded without the BATT_TYPE setting above, the application assumes a non-rechargeable battery on VBAT, and disable VBATLI pin (with the exception of BLE_harv_adv app, in which the battery type is determined by the DIP switch setting on the EVK). During evaluation, it is possible to regain access to VBATLI by following the procedure below:

1. power the device from VBAT.
2. reprogram the device with the correct BATT_TYPE setting.
3. remove the power at VBAT.
4. apply power at VBATLI and proceed with evaluation.

5.2 Enable VBATLI for ATM33e

To enable high-voltage battery or storage device and charging with energy harvesting in the SDK examples, the user can insert the following compilation flags in the corresponding sections of the makefile (ie. platform/atm33/ATM33xx-5/examples/BLE_adv/makefile), build the application and load it to the target device.

1) Enable VBATLI pin for high-voltage battery or storage device. Please refer to Enable VBATLI for ATM33 section for the procedure.

```
CFLAGS += \
    -DBATT_TYPE=BATT_TYPE_LI_ION \
    -DBATT_LEVEL=BATT_LEVEL_GT_1P8V \
    -DVDDIO_TYPE=VDDIO_TYPE_INTERNAL \
```

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2) Add the driver for battery level monitor and charging path control. For additional variations associated with battery monitoring, please refer to Battery Monitoring for VBATLI section.

```
DRIVERS += \
    batt_model_liion \
```

3) Enable VBATLI charging with RF or PV harvesting.
    a) Enable charging with RF harvesting.

```
CFLAGS += \
    -DCFG_RF_HARV \
    -DVSTORE_MAX=VSTORE_MAX_3P3V \
```

    b) Enable charging with PV harvesting, i.e. with a Panasonic PV cell AM-1454 that has optimized power between 1.7 V - 2.0 V at VHARV. For a complete list of supported VHARV range for different PV cells, please refer to the **ATM33/e Series Reference Manual, Harvesting type** section.

```
CFLAGS += \
    -DCFG_NONRF_HARV \
    -DVHARV_RANGE=VHARV_RANGE_1P7V_2P0V \
    -DVSTORE_MAX=VSTORE_MAX_3P3V \
```

4) Modify the VBATLI charging related thresholds, refer to Table 4 for detailed descriptions. The following thresholds are examples for different battery types. Please check the maximum spec for your device.

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a) Hybrid supercapacitor (HSC): max voltage spec is 3.8 V.

```
CFLAGS += \
    -DVBAT_OV=3.8f \
    -DVBAT_OV_HYS=3.75f \
    -DVBAT_BOOST=3.2f \
    -DVBAT_BOOST_HYS=3.15f \
```

b) Li-ion battery: max voltage spec is 4.2 V.

```
CFLAGS += \
    -DVBAT_OV=4.2f \
    -DVBAT_OV_HYS=4.15f \
    -DVBAT_BOOST=3.2f \
    -DVBAT_BOOST_HYS=3.15f \
```

c) LiFePO4 battery (LFP): max voltage spec is 3.6 V

```
CFLAGS += \
    -DVBAT_OV=3.6f \
    -DVBAT_OV_HYS=3.55f \
    -DVBAT_BOOST=3.2f \
    -DVBAT_BOOST_HYS=3.15f \
```

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5.3 Modify VBATLI Brownout Levels

The brownout driver is enabled by default. To modify the thresholds in the SDK examples, the user can add the following compilation flags in the corresponding sections of the makefile (ie. platform/atm33/ATM33xx-5/examples/BLE_adv/makefile), build the application and load it to the target device.

```
CFLAGS += \
    -DBRWNOUT_THR_VBATLI=2.8f \
    -DWAKEUP_THR_VBATLI=26 \
```

Note **BRWNOUT_THR_VBATLI** is defined as the actual voltage threshold, while **WAKEUP_THR_VBATLI** is an index that can be converted to VBATLI voltage level through this equation:

VBATLI = 1.8 + 0.05*X where 14 <= X <= 31

To disable the brownout driver during evaluation, the user can add this setting in the makefile or in the command line: NO_BROWNOUT=1

5.4 Battery Monitoring for VBATLI

The BLE_harv_adv example also demonstrates the use of monitoring the battery level periodically.

The following CFLAGS in the makefile can be used to configure the behavior:

```
CFLAGS += \
    -DCFG_APP_FEATURE=APP_BATT_AUTO_TIMER_BIT \
    -DAPP_BATT_LOW_POLL_TIME_CS=100 \
    -DAPP_BATT_HIGH_POLL_TIME_CS=600 \
    -DBATT_MEASUREMENT_CNT_MAX=1 \
```

- CFG_APP_FEATURE=APP_BATT_AUTO_TIMER_BIT activates the periodic monitoring implemented in the battery management library lib/app_batt/app_batt.c

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- APP_BATT_LOW_BATTERY_PERCENTAGE: Threshold below which the battery is considered to be low.
- APP_BATT_LOW_POLL_TIME_CS: Polling interval when battery is running low. Unit in centisecond.
- APP_BATT_HIGH_POLL_TIME_CS: Polling interval when battery is not running low. Unit in centisecond.
- BATT_MEASUREMENT_CNT_MAX: Number of times the battery check is performed before the chip goes into the configured low power state.

Reference Documents

Title	Document Number
ATM33e Series Energy Harvesting Application Note	ATM33e-ANEH
ATM33/e Energy Harvesting EVK Quick Start Guide	ATM33_e-QSGEH
ATN33/e EVK Power Consumption Evaluation User Guide	ATM33_e-UGPCE
ATM33/e Series Evaluation Kit User Guide	ATM33_e-UGEVK
ATM33/e Series Reference Manual	ATM33_e-RM
SDK User Guide	ATM-UGSDK

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Revision History

Date	Version	Description
March 10, 2023	0.52	Updated Overview, Hardware and Software Requirements, Supported SDK, Power Management and Harvesting, VBATLI Supply Path Operation for ATM33 and ATM33e, VBATLI System Brownout for ATM33 and ATM33e, Table 5 - VBATLI Brownout Related Thresholds and Descriptions, Enable VBATLI for ATM33, Enable VBATLI for ATM33e, Modify VBATLI Brownout Levels, Battery Monitoring for VBATLI sections, Figure 1 - Block Diagram for ATM33 Operating from VBATLI, Figure 2 - Block Diagram for ATM33e Operating from VBATLI, Figure 6 - VBATLI Charging Path 1: Charging Through An External Charge Pump for ATM33e, Figure 7 - VBATLI Charging Path 2: Charging Through On-chip Path of ATM33e, Figure 8 - Battery Charging Operations for Different Battery Levels. Added Battery Monitoring for VBATLI section. Renamed to ATM33/e Support for Storage Devices Application Notes.
August 16, 2022	0.51	Updated various sections of the document.
July 20, 2022	0.50	Initial version created.

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